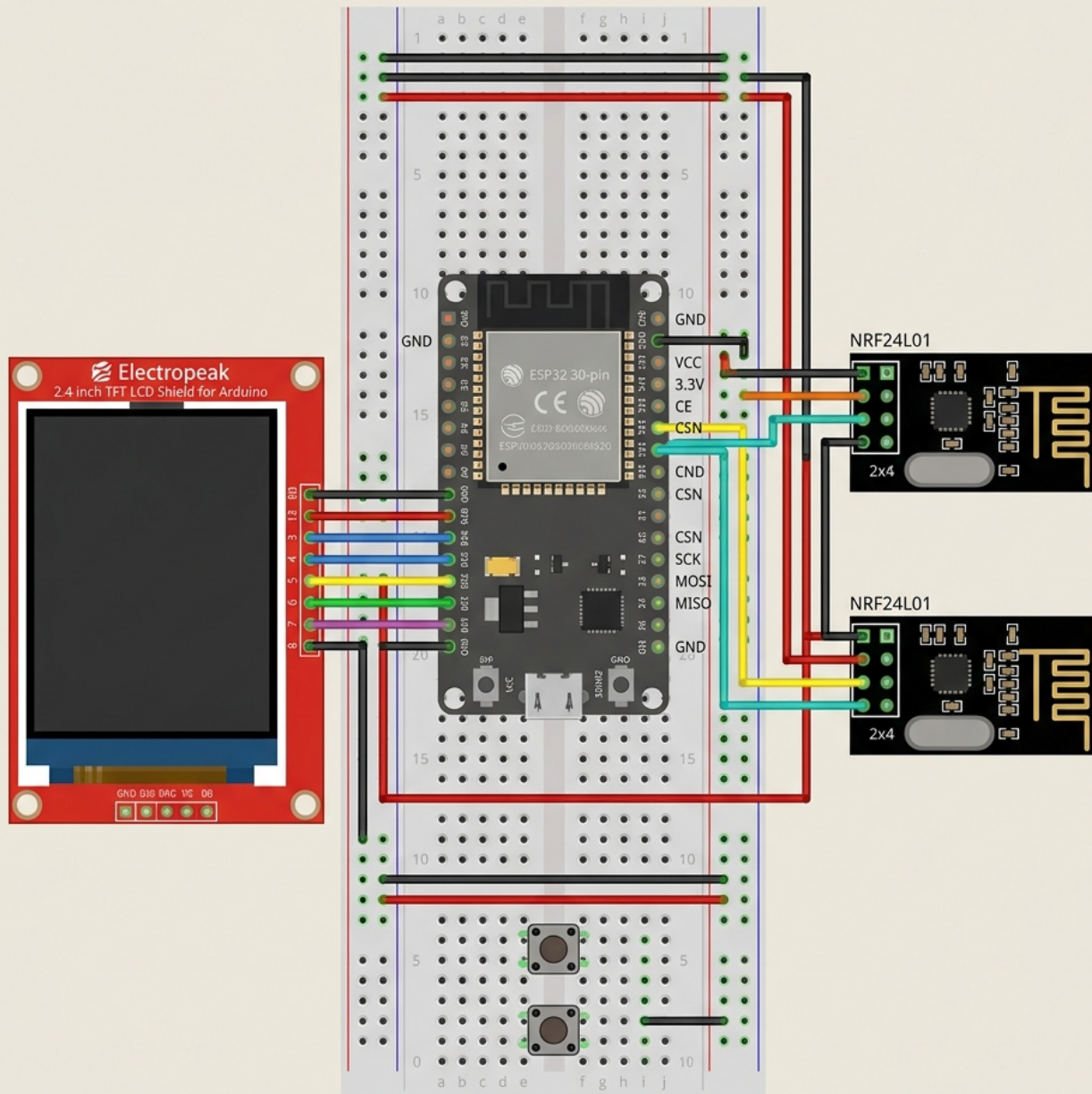


Master Hardware Pinout, RF Transceiver & Wiring Diagram Guide

This device design and documentation are strictly for educational and authorized penetration testing in controlled environments. Operating RF jammers or packet injection without authorization is illegal.



2. ESP32 Pin Allocation Overview

ESP32 30-PIN DEV BOARD LAYOUT			
Radio 1 CE	GPIO 22	GPIO 21	Radio 1 CSN
Radio 2 CE	GPIO 16	GPIO 15	Radio 2 CSN
Touch T_CS	GPIO 27	GPIO 4	TFT DC / RS
TFT Backlight	GPIO 32	GPIO 5	TFT CS
TFT Reset	GPIO 33	GPIO 18	SPI SCK (Clock)
Physical ON Btn	GPIO 12	GPIO 19	SPI MISO
Physical OFF Btn	GPIO 14	GPIO 23	SPI MOSI
Power Rail	3.3V	GND	Ground Rail

3. Hardware Interfaces (VSPI & ESP32 Wi-Fi Radio)

Signal / Interface	ESP32 Pin	Connected Modules / Mode	Type
SCK (Clock)	GPIO 18	Radio 1, Radio 2, TFT SCK, Touch T_CLK	Shared SPI
MISO (Master In)	GPIO 19	Radio 1, Radio 2, TFT SDO, Touch T_DO	Shared SPI
MOSI (Master Out)	GPIO 23	Radio 1, Radio 2, TFT SDI, Touch T_DIN	Shared SPI
Wi-Fi 802.11 b/g/n	Internal RF	AP Scanning, Promiscuous Sniffer, Web Portal	Internal RF

4. Radio Transceiver Modules (NRF24L01)

Radio #1 (Lower Channels 2402-2440 MHz):

NRF24 #1 Pin	ESP32 Pin	Function	Signal Category
VCC / GND	3.3V / GND	Power Rail (+3.3V Max)	Power / Ground
CE	GPIO 22	Chip Enable	Dedicated Control
CSN	GPIO 21	SPI Chip Select	Dedicated SPI CS
SCK / MOSI / MISO	GPIO 18 / 23 / 19	Hardware SPI Bus	Shared SPI

Radio #2 (Upper Channels 2441-2480 MHz):

NRF24 #2 Pin	ESP32 Pin	Function	Signal Category
VCC / GND	3.3V / GND	Power Rail (+3.3V Max)	Power / Ground
CE	GPIO 16	Chip Enable	Dedicated Control
CSN	GPIO 15	SPI Chip Select	Dedicated SPI CS
SCK / MOSI / MISO	GPIO 18 / 23 / 19	Hardware SPI Bus	Shared SPI

5. 2.4" TFT LCD Shield (8-Bit Parallel Bus)

Shield Pin	ESP32 Pin	Function	Signal Category
5V / 3V3 / GND	5V / 3.3V / GND	Display Power & Ground	Power / Ground
LCD_RST	GPIO 33	Hardware Reset	Dedicated Control
LCD_CS	GPIO 5	Chip Select	Dedicated Control
LCD_RS	GPIO 4	Register / Command Select	Dedicated Control
LCD_WR	GPIO 2	Write Enable	Dedicated Control
LCD_RD	3.3V (Tie HIGH)	Read Enable (Not Used)	Power
LCD_D0	GPIO 12	Data Bit 0	8-Bit Parallel
LCD_D1	GPIO 13	Data Bit 1	8-Bit Parallel
LCD_D2	GPIO 26	Data Bit 2	8-Bit Parallel
LCD_D3	GPIO 25	Data Bit 3	8-Bit Parallel
LCD_D4	GPIO 17	Data Bit 4	8-Bit Parallel
LCD_D5	GPIO 27	Data Bit 5	8-Bit Parallel
LCD_D6	GPIO 14	Data Bit 6	8-Bit Parallel
LCD_D7	GPIO 32	Data Bit 7	8-Bit Parallel
SD_SS (CS)	GPIO 13	SD Card Chip Select	SPI CS
SD_DI (MOSI)	GPIO 23	SD SPI Data In	Shared SPI
SD_DO (MISO)	GPIO 19	SD SPI Data Out	Shared SPI
SD_SCK (SCK)	GPIO 18	SD SPI Clock	Shared SPI

6. Hardware Pushbuttons & System Controls

Button / LED	ESP32 Pin	Wiring Connection	Mode
TURN ON Button	GPIO 12	One leg to GPIO 12, other leg to GND	INPUT_PULLUP
TURN OFF Button	GPIO 14	One leg to GPIO 14, other leg to GND	INPUT_PULLUP
BOOT Button	GPIO 0	Built-in BOOT pushbutton on ESP32 board	Internal
Status LED	GPIO 2	Built-in LED on ESP32 board	OUTPUT

7. Power Supply & Capacitor Recommendations

- 1. Built-in PCB Antenna Modules: Draw ~12mA each. NO CAPACITORS REQUIRED! Power directly from ESP32 3.3V pin.
- 2. High-Power PA+LNA Modules: Draw ~115mA peak each. Requires 10uF-47uF capacitors across VCC & GND to buffer current spikes.